shift from semi-autonomous reconnaissance systems to fully networked, satellite-linked, and combat-capable platforms. Building upon the technological groundwork laid by second-generation UAVs, such as the integration of GPS navigation, telemetry, and digital flight control, this generation focused on global connectivity, endurance, precision, and mission adaptability.

The roots of third-generation drones can be traced back to the late 1990s, when advancements in satellite communication (SATCOM) and miniaturized electronics made it feasible to operate UAVs beyond visual and radio line-of-sight. The RQ-1 Predator, developed by General Atomics Aeronautical Systems (GA-ASI) and first introduced in 1995, became the foundation of this new generation. Initially designed as a surveillance drone, the Predator was later equipped with AGM-114 Hellfire missiles, transforming it into the first armed UAV capable of conducting precision strikes. Its operational success in the War on Terror, particularly in Afghanistan and Iraq, demonstrated the effectiveness of drones as persistent aerial assets capable of both surveillance and combat roles.

By the early 2000s, UAV development had accelerated rapidly across several countries. The RQ-4 Global Hawk, developed by Northrop Grumman, represented a leap forward in high-altitude long-endurance (HALE) UAV design. Capable of flying above 60,000 feet for over 36 hours, it provided near-real-time, high-resolution imagery across vast regions. Its combination of synthetic aperture radar (SAR), electro-optical (EO), and infrared (IR) sensors made it invaluable for strategic reconnaissance and intelligence gathering. The Global Hawk’s ability to autonomously traverse continents and transmit live data via satellite marked a defining moment in UAV autonomy and global reach.